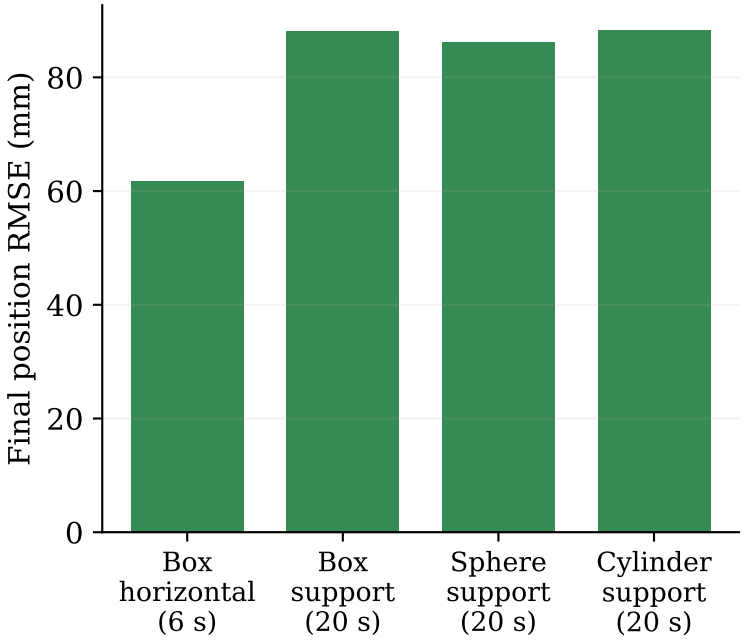

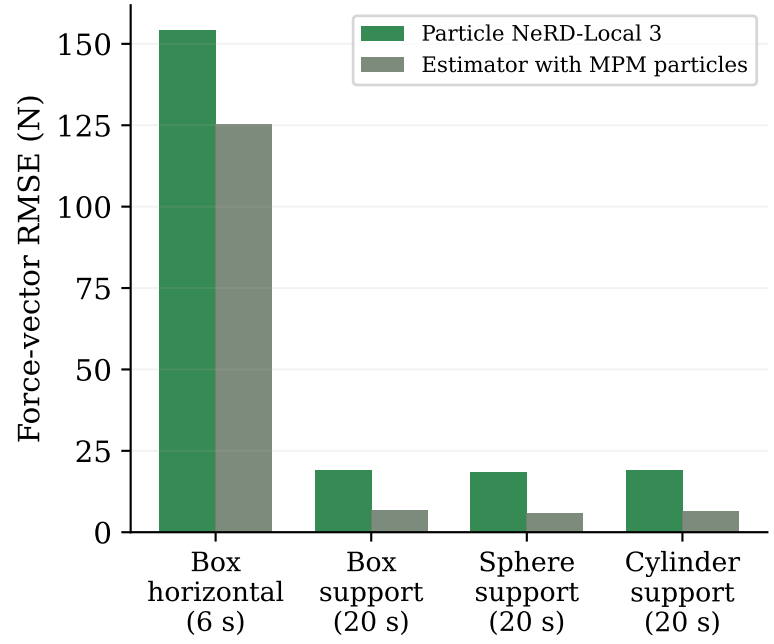

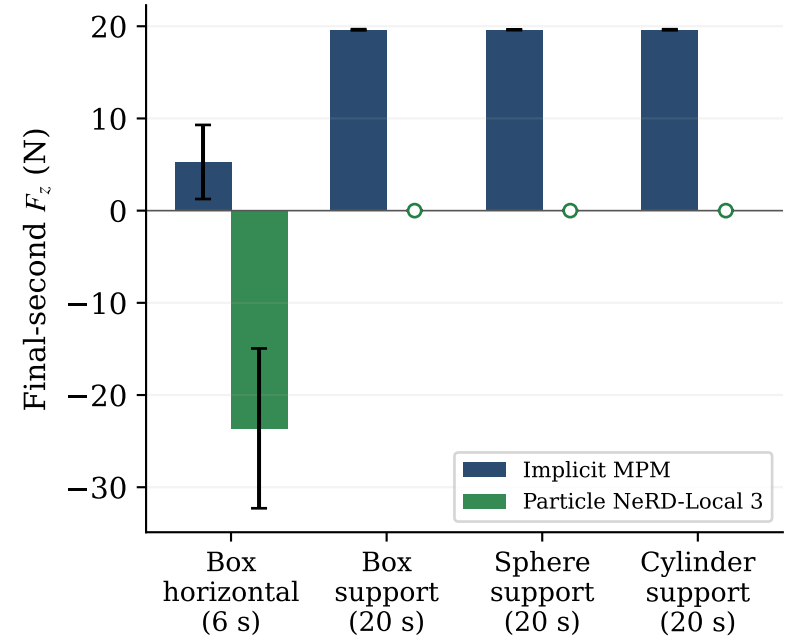

Figure 6: Frozen object-model checkpoint 4,500. Left: terminal particle-position RMSE. Center: force-vector RMSE without division by three. Right: mean Fz during the final second, with temporal population standard deviation. Open circles mark measured zeros. Whiskers describe temporal variation in one recording, not repeated-run uncertainty. Horizontal motion lasts 6 s; support tests last 20 s. Body poses are replayed from MPM and do not respond to estimated force.